Name
Date
Course

Final Test (Part I)

I) This exam covers the first half of the course (everything up to test II).
II) Write your name, the date, and the course title (i.e. Precalculus).
III) Show all work.
IV) Draw a rectangle around your final answer for each problem.
V) You may use a calculator for counting problems only, not for graphing.
VI) Each problem is worth four points for a total of 100 points.

① If the 3rd term of an arithmetic sequence is −15, and the 9th term is 33, what is the 10th term?

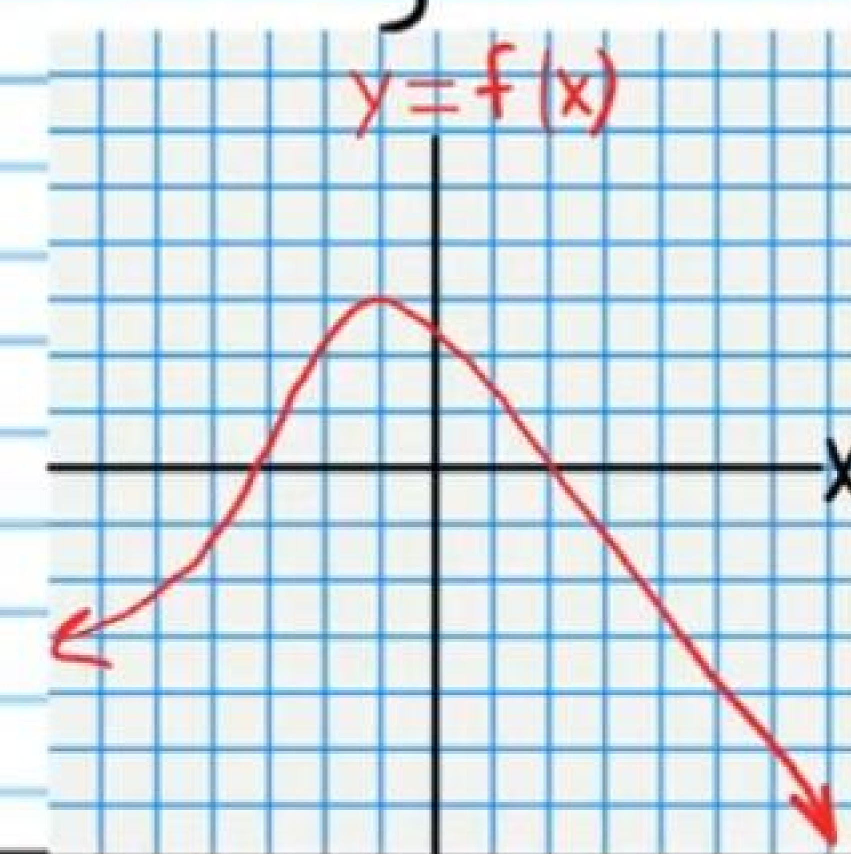
② How many different ways can 7 people be lined up?

③ It is Tommy's birthday. His mom said that he can pick 4 of his 6 friends to play video games with for the whole day. His mom also said that he can choose 3 new video games out of the 8 that he likes at the game store, 3 places to order food out of his 5 favorite restaurants, 4 Ninja Turtle action figures out of the 9 available, and 7 football cards out of the 9 that he likes. How many outcome variations are possible?

④ The graph of a function is written below. Write the intervals where the function is increasing or decreasing.

Increasing:

Decreasing:

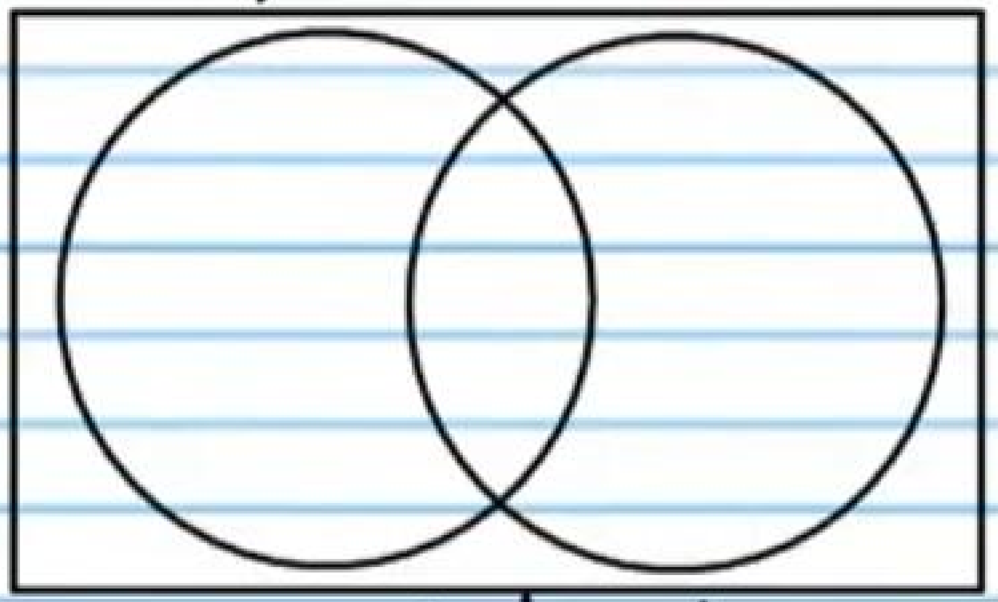


⑤ A school requires every student to choose at least one area of science to study. There are 400 students at the school. 177 of them choose either biology or chemistry. 116 of them choose biology and 96 choose chemistry.

i) How many students choose both chemistry and biology?

ii) How many students choose biology but not chemistry?

iii) What is the probability that a randomly selected student will choose both biology and chemistry?



iv) What is the probability that a randomly selected student chooses biology but not chemistry?

⑥ The probability that Linda will not be accepted to her college of choice is 64%. What is the probability that she will be accepted?

⑦ Use the Binomial Theorem to expand the following expression.

$(4x+y^2)^3$

⑧ i) Write an equation of the n^{th} term of the sequence below in recursive form (show the value of the first term).

$2, 1\frac{1}{3}, \frac{2}{3}, 0, \ldots$

ii) What is the 50^{th} term?

9. Find the value of the expression below.

$$\sum_{n=1}^{3} n^2+10$$

10. Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

Zeros: $2+4i, -1$ $\quad LC=1 \quad D=3$

11. i) Find the sum of the first 24 terms of the sequence below.

100, 25, 12.5, 3.125, ...

ii) What is the sum of all of the terms?

⑫ Divide.

$(2x^6 - 6x) \div (1 + x^3)$

⑬ Find the real zeros of the following polynomial function using the grouping method. It is not necessary in this problem to find the imaginary zeros.

$f(x) = x^4 - 2x^3 - 125x + 250$

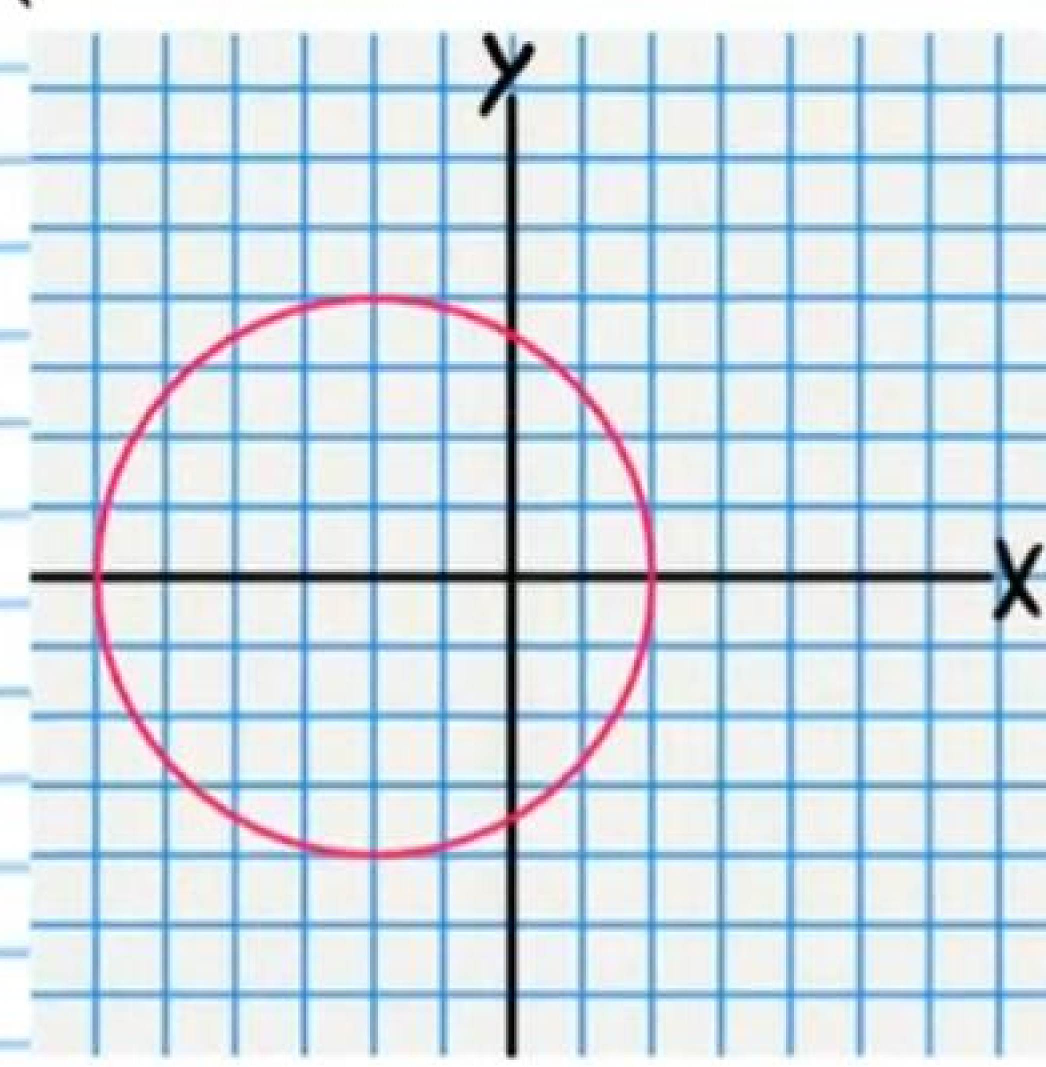
⑭ Write the first three terms of the sequence written below. Assume that the first term is produced when $n=1$.

$\{n^2\}$

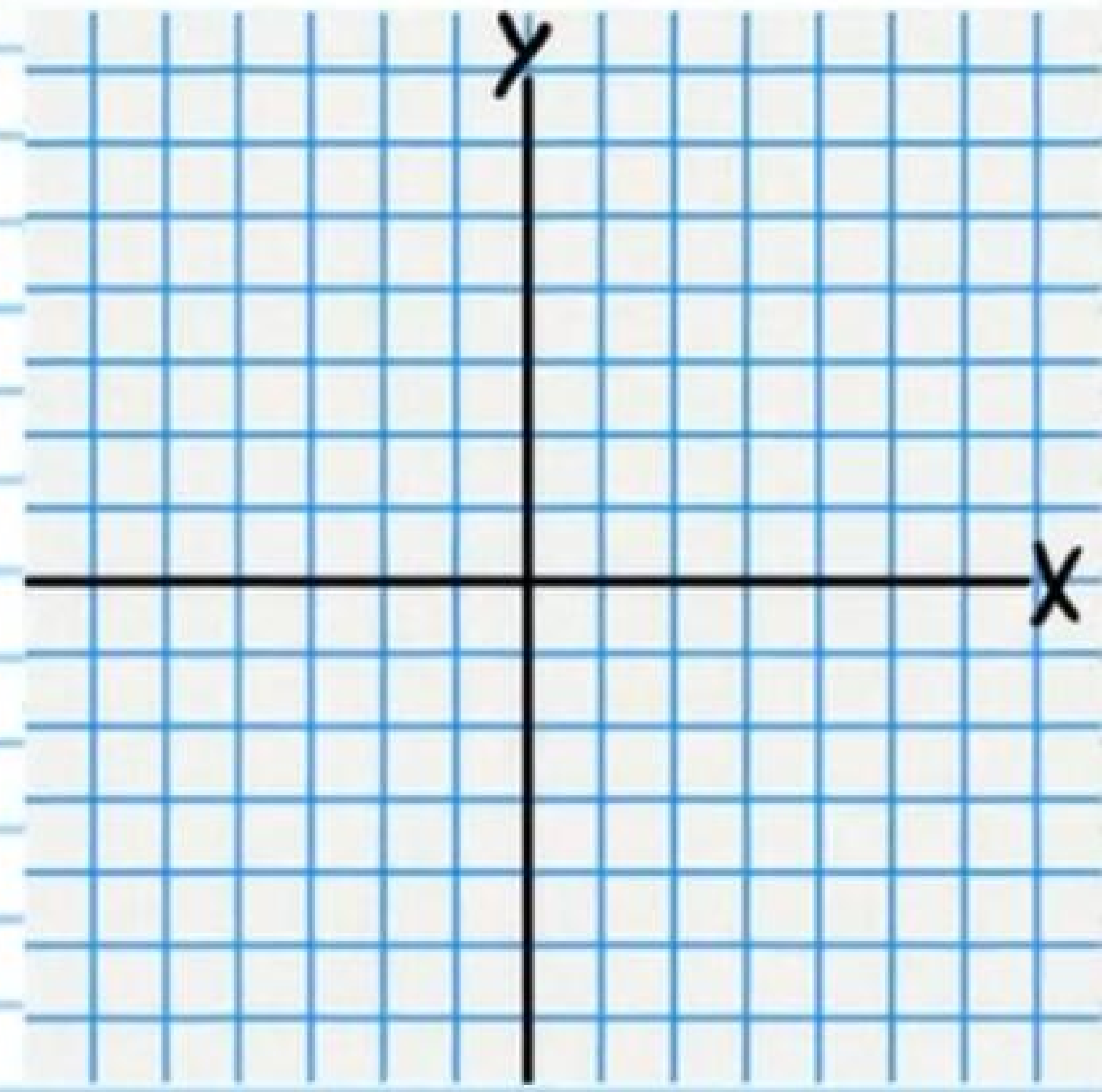
⑮ Indicate the end behavior of the function below.

$f(x) = x^3 - x^2 + x + 1$

⑯ Write an equation of the graph.



⑰ Find an equation of the hyperbola with the following characteristics: Center at $(-1,2)$ Focus at $(3,2)$, Vertex at $(1,2)$. Then graph the equation.

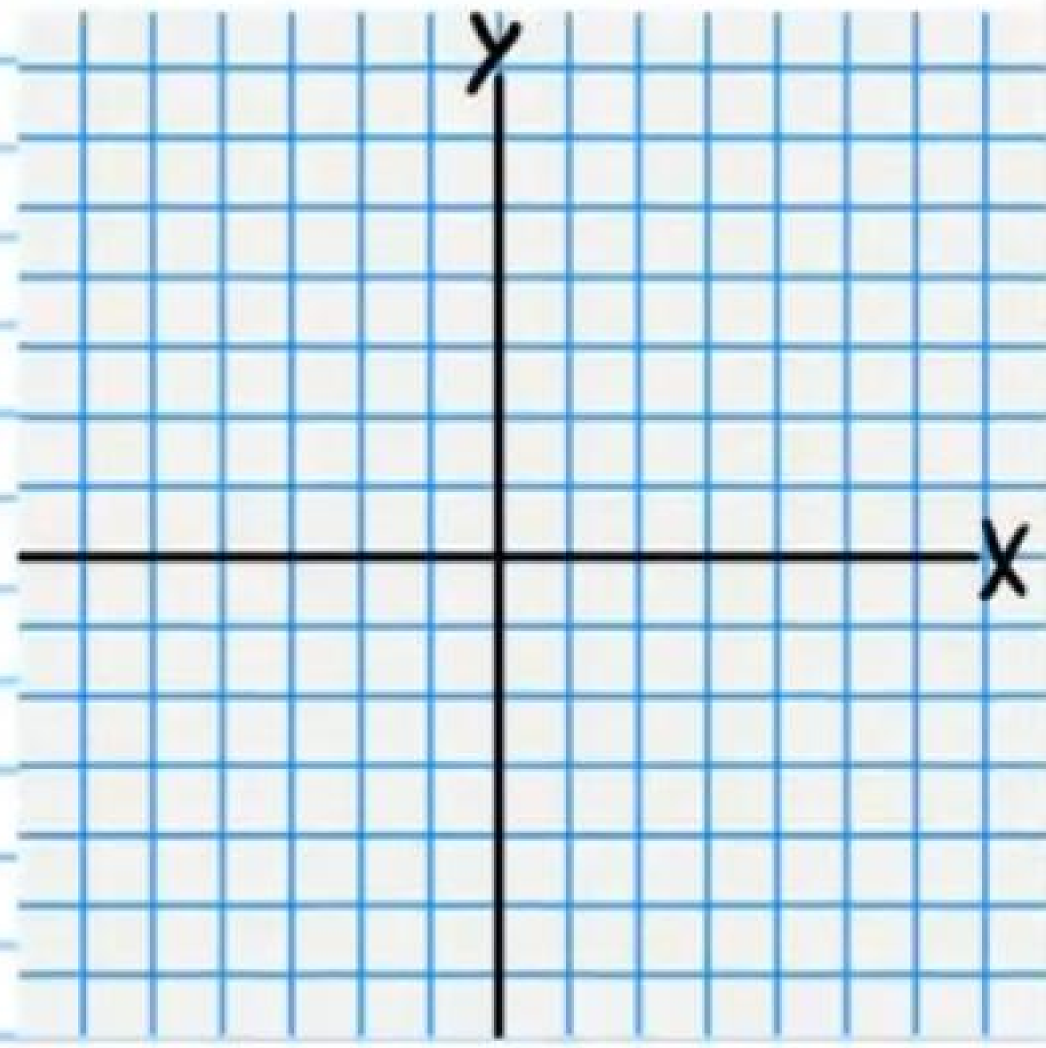


(18) Solve. Write the solutions graphically and in interval notation.

$$\frac{x^4-3x^2-70}{3x^3-3x^2+x-1}<0$$

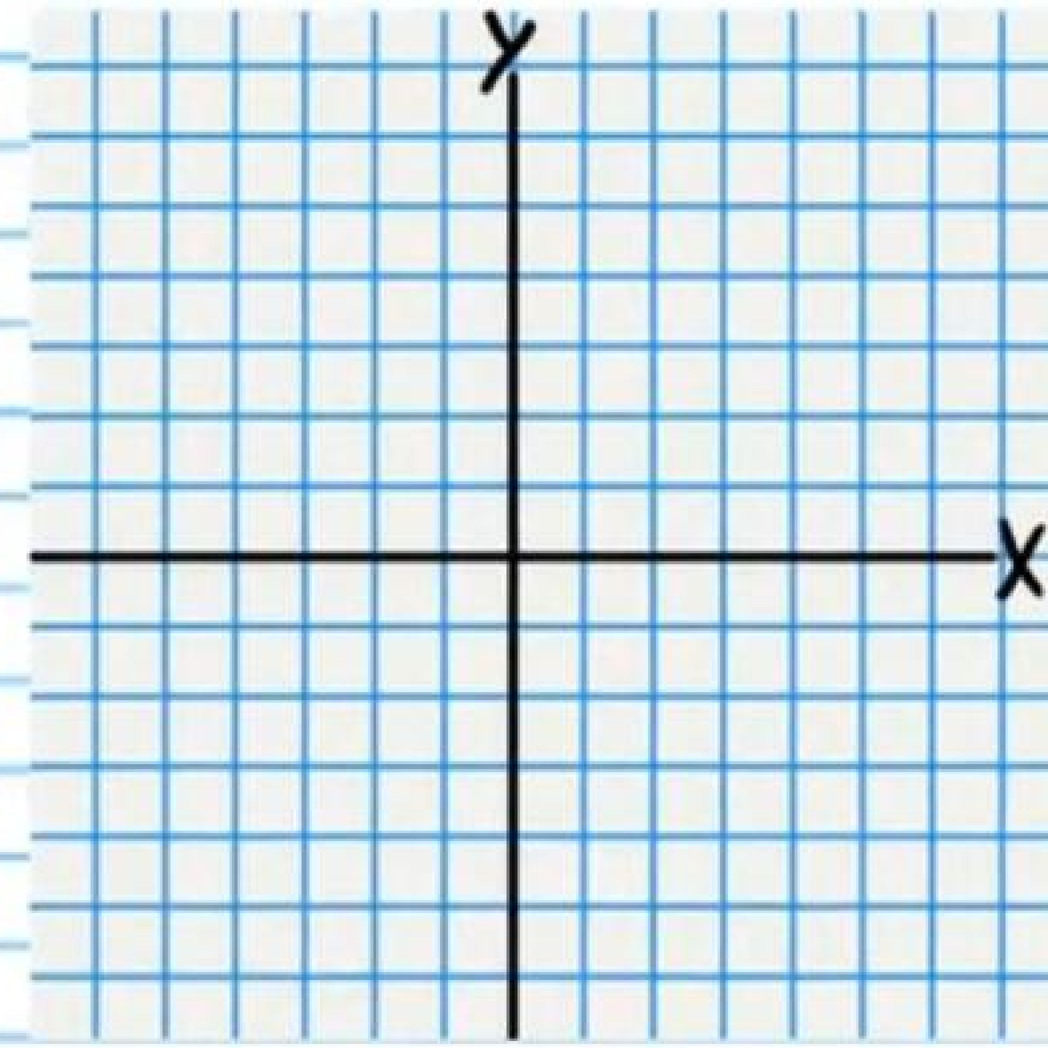
(19) Graph the equation below.

$$4x^2+y^2+32x+4y+52=0$$



(20) Graph the equation below.

$y^2 - x^2 + 2y + 2x - 9 = 0$



(21) Solve. Write the solutions graphically and in interval notation.

$x^3 \geq 9x$

(22) Find the zeros of the function below (real and imaginary).

$f(x) = x^3 + 3x + 4$

(23) Graph the following function. Draw asymptotes as dotted lines and write their equations.

$f(x) = \frac{1}{x^2+8x+12}$

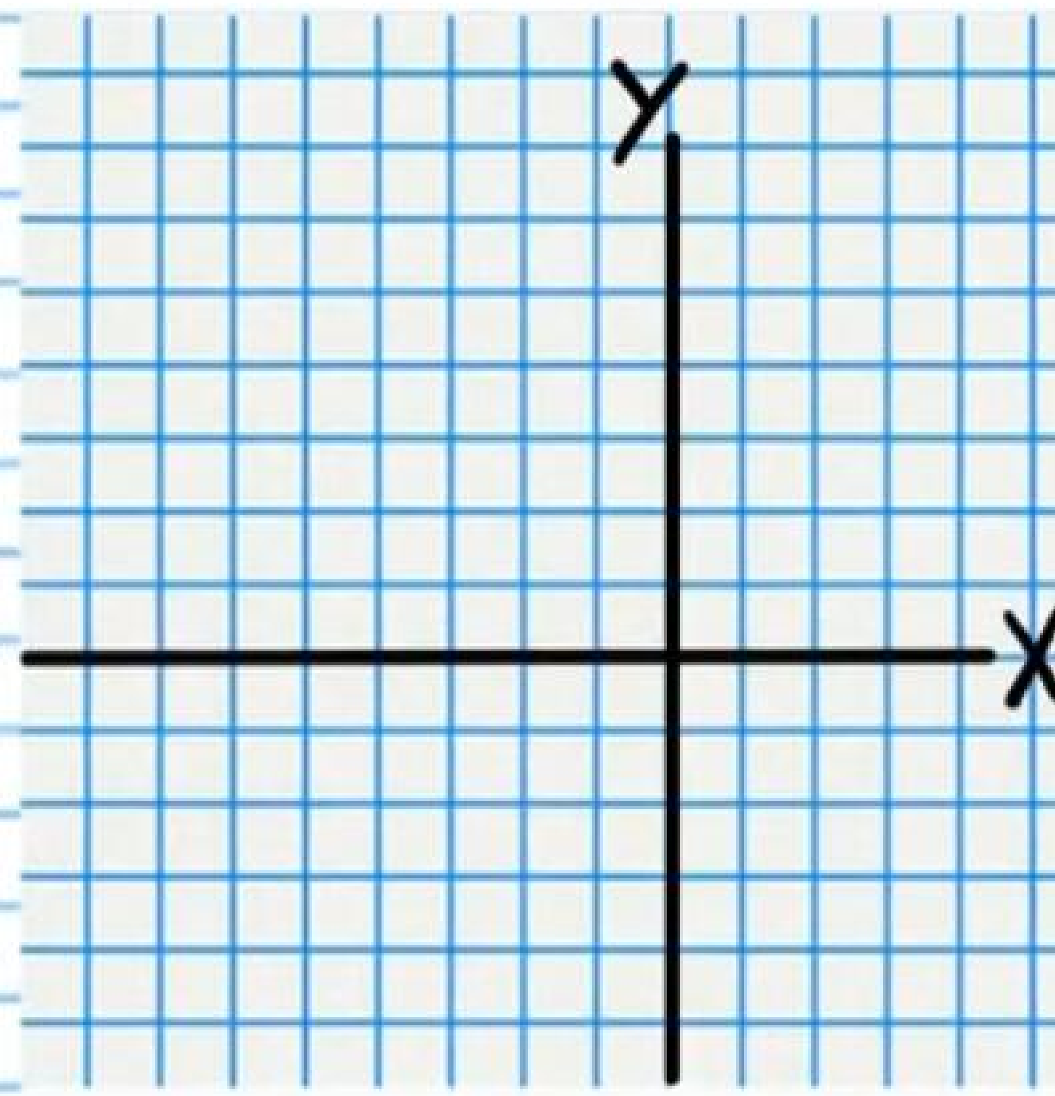
Vertical Asymptote Equation(s)

Horizontal Asymptote Equation(s)

Sign Chart

Y-Intercept

X-Intercept



(24) For the Function, Find the information requested.
It is not necessary to graph the function.

$$f(x) = \frac{x^4-16}{x^2+2x-8}$$

Asymptote Curve

Y-Intercept

X-Intercept(s)

Vertical Asymptote Equation(s)

Hole(s)

(25) State the maximum number of turning points of the graph of the function below.

$$f(x) = 9x^7-4x^6+2x^5+x^4-x^3+2x^2+83x-14$$

Answers

① $\boxed{41}$

② $\boxed{5{,}040 \text{ ways}}$

③ $\boxed{38{,}102{,}400 \text{ outcome variations}}$

④ Increasing: $(-7,-1)$

Decreasing: $(-1,7)$

⑤ i) $\boxed{35 \text{ students}}$

ii) $\boxed{81 \text{ students}}$

iii) $\boxed{\frac{7}{80}}$

iv) $\boxed{\frac{81}{400}}$

⑥ $\boxed{36\%}$

⑦ $\boxed{64x^3+48x^2y^2+12xy^4+y^6}$

⑧ i) $\boxed{A_1=2,\ A_n=A_{n-1}-\frac{2}{3}}$

ii) $\boxed{a_{50}=-\frac{92}{3}}$

⑨ $\boxed{44}$

⑩ $\boxed{x^3-3x^2+16x+20}$

⑪ i) $\boxed{S_{24}=400\left(\frac{1-\left(\frac{1}{4}\right)^{24}}{3}\right)}$

or $\boxed{S_{24}=\frac{400}{3}\left(1-\left(\frac{1}{4}\right)^{24}\right)}$

ii) $\boxed{\frac{400}{3}}$ or $\boxed{133\frac{1}{3}}$

⑫ $\boxed{2x^3-2+\frac{-6x+2}{x^3+1}}$

⑬ $\boxed{x=5,\ x=2}$

⑭ $\boxed{1,4,9}$

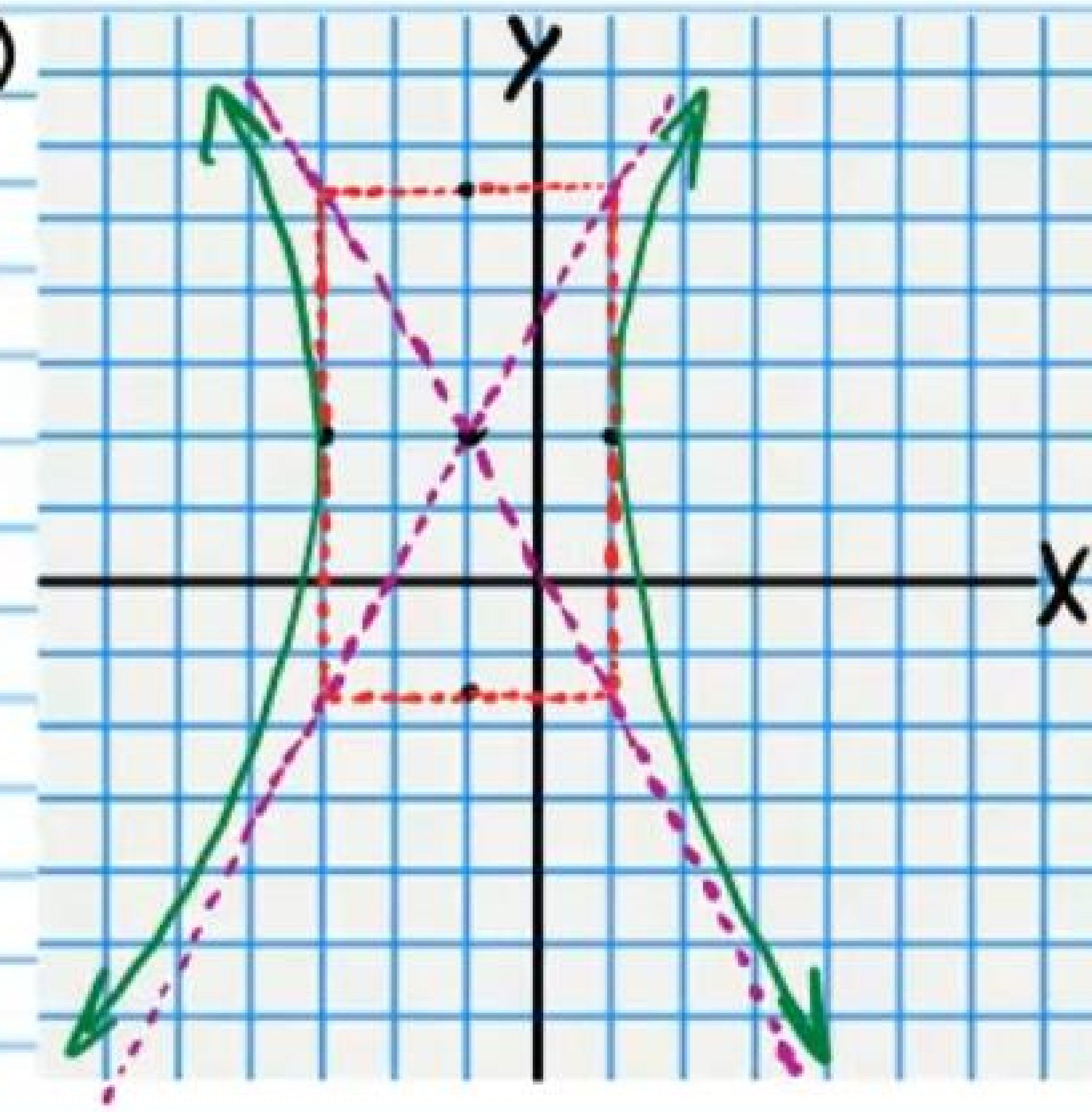
⑮

As $x\to\infty,\ f(x)\to\infty$

As $x\to-\infty,\ f(x)\to-\infty$

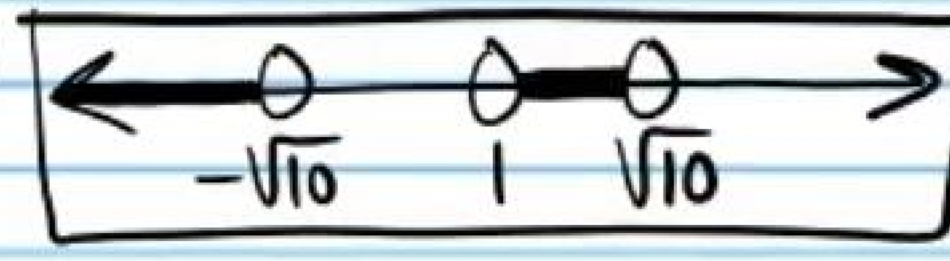
⑯ $\boxed{(x+2)^2+y^2=16}$

⑰



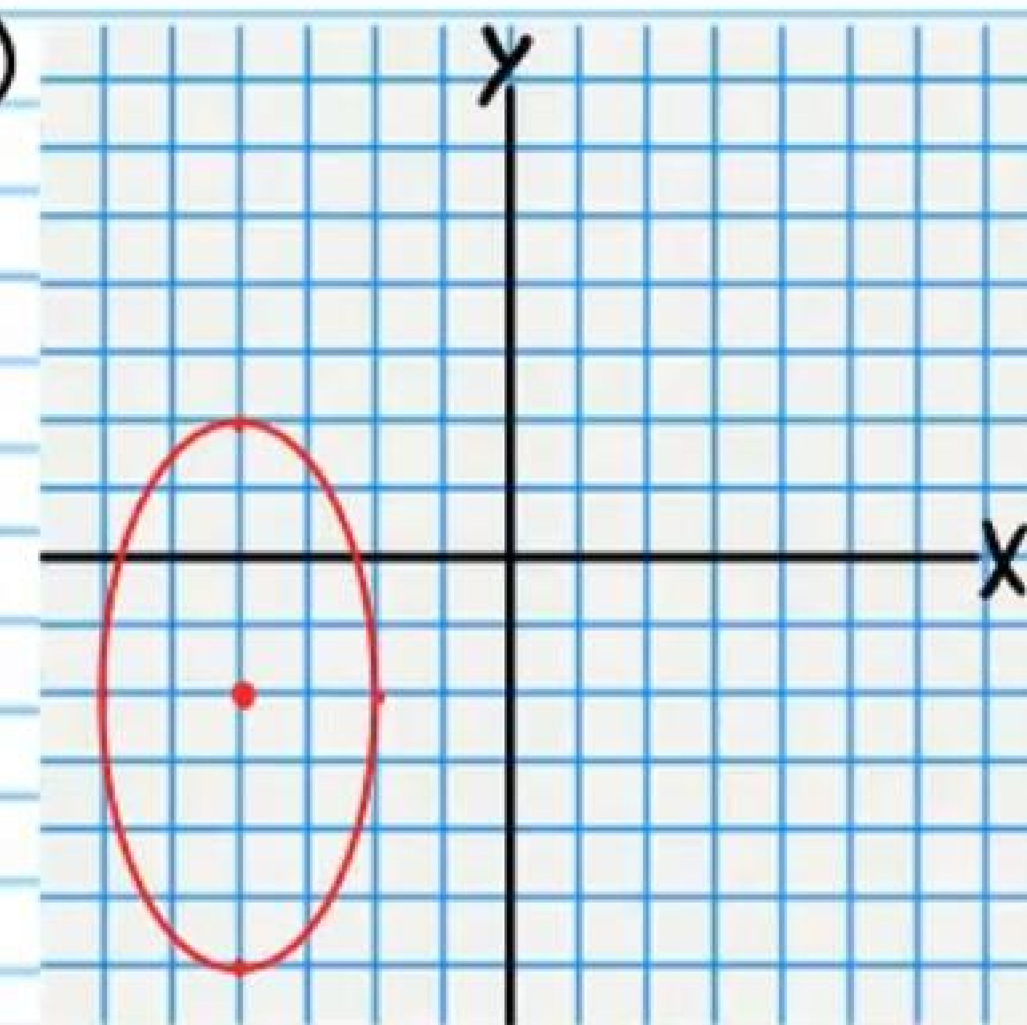
$\boxed{\frac{(x+1)^2}{4}-\frac{(y-2)^2}{12}=1}$

⑱

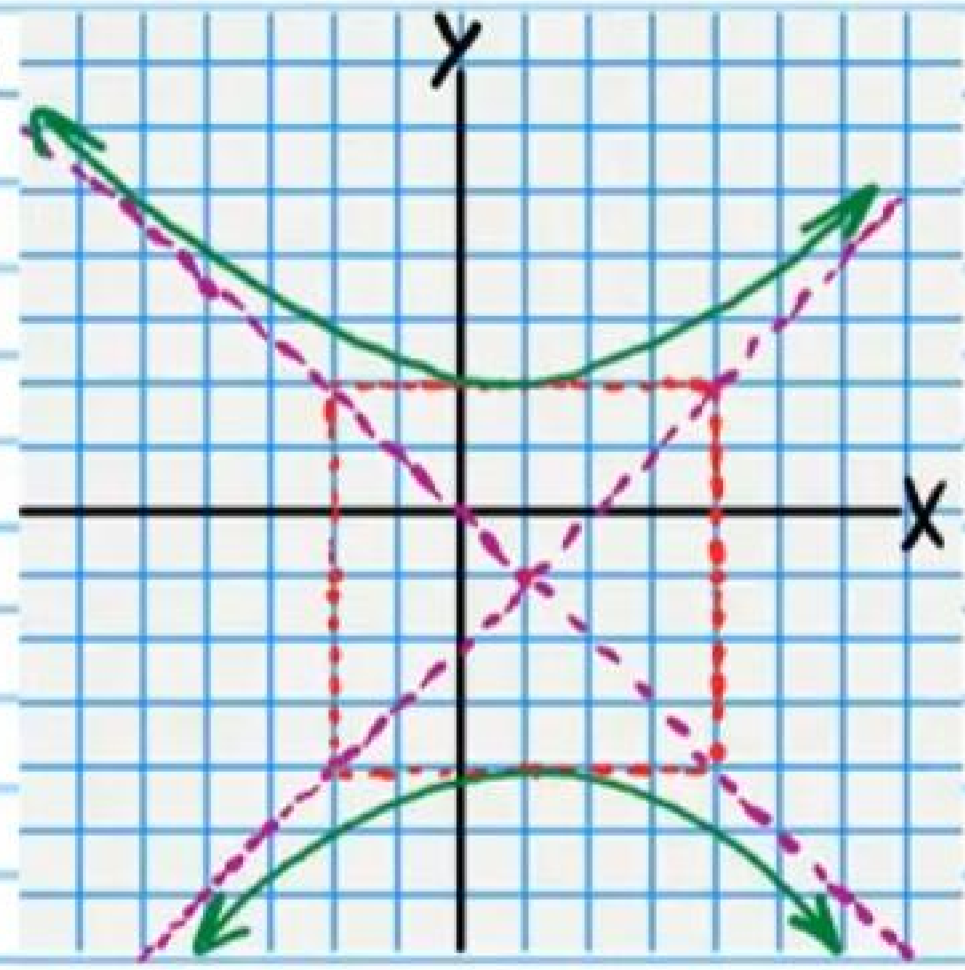


$\boxed{(-\infty,-\sqrt{10})\cup(1,\sqrt{10})}$

⑲



(20)



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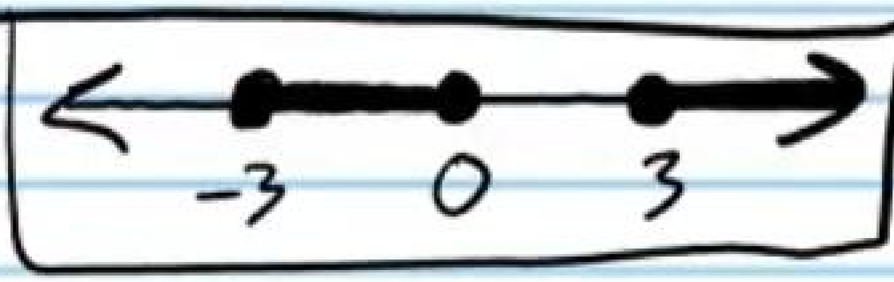
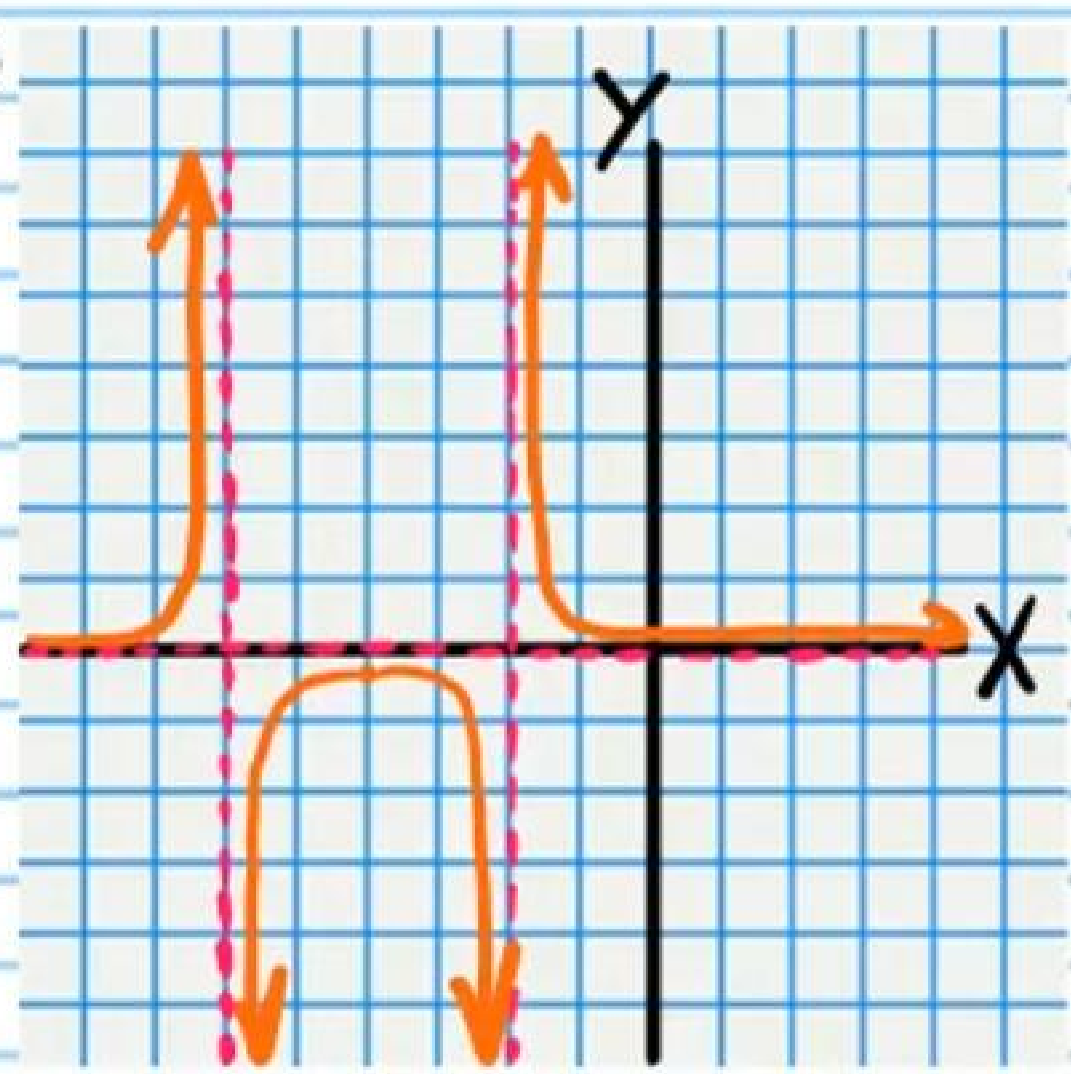


$[-3,0] \cup [3,\infty)$

(22) $x=-1,\ x=\frac{1+i\sqrt{15}}{2},\ x=\frac{1-i\sqrt{15}}{2}$

(23)



$y=0,\ x=-2,\ x=-6$

(24) $y=x^2-2x+12$

VA: $x=-4$

Hole: $x=2$

X int: $(-2,0)$

Y int: $(0,2)$

(25) 6